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DATA DESCRIPTOR

AS-OCT dataset with anatomical structure segmentation and scleral spur localization in cataract and glaucoma

Jiongning Zhao^{1,2,5}, Huihui Fang^{2,5}, Yuetong Yang^{2,3}, Xinyu Fu^{2,3}, Jingru Deng¹, Jinghe Yu¹, Xingying Yan¹, Xinya Hu⁴, Xiaoqing Wang⁴, Yuting Hu⁴, Di Gong⁴, Zhe Zhang⁴, Wei Chi⁴✉, Weihua Yang⁴✉ & Yanwu Xu^{1,2}

Anterior Segment Optical Coherence Tomography (AS-OCT) provides high-resolution, non-invasive visualization of the anterior eye and is widely used for clinical diagnosis and surgical planning. However, automated analysis of AS-OCT images remains limited by the lack of publicly available datasets with comprehensive anatomical annotations and disease labels. Here we present the AS-OCT Multi-Structure and Disease Classification Dataset (ASOCT-MSDC), a curated dataset designed for anatomical structure segmentation and disease classification. The dataset contains 1106 AS-OCT images from 1106 eyes of 627 individuals across four clinical categories: normal, glaucoma, cataract, and glaucoma-cataract comorbidity. Each image underwent stringent quality assessment, and three standardized image quality scores—eyelid obscuration, black-line anomaly, and central light artifact—are provided as metadata. Expert-validated annotations include segmentation masks for the anterior chamber, iris, lens, and nucleus, together with scleral spur localization points. By combining anatomical annotations, disease labels, and image quality metadata, ASOCT-MSDC supports research on anatomical segmentation, disease classification, and image quality assessment in anterior segment OCT images.

Background & Summary

Anterior Segment Optical Coherence Tomography (AS-OCT) has significantly advanced ophthalmic imaging by providing high-resolution, in-vivo visualization of the anterior chamber¹. This non-invasive technology is integral to modern clinical practice for quantifying key anatomical parameters for both cataract surgery and glaucoma management², such as anterior chamber depth (ACD), lens vault (LV), and angle opening distance (AOD)^{3,4}, overcoming the inherent subjectivity and qualitative limitations of traditional gonioscopy⁵. Recent systematic reviews and meta-analyses have demonstrated that deep learning-based AS-OCT image analysis has achieved expert-level accuracy in detecting angle closure and identifying relevant anatomical landmarks, providing robust support for personalized risk stratification, longitudinal monitoring of disease progression, and treatment planning⁶.

However, the translation of this rich imaging data into actionable clinical insights is hindered by a reliance on manual analysis. This manual process is not only time-consuming and expertise-dependent, creating a significant bottleneck in busy clinics⁷, but it is also fundamentally subjective. Even among experts, manual measurements are prone to considerable inter-observer variability, which can compromise diagnostic consistency and surgical planning⁸. This challenge is exacerbated by a global shortage of ophthalmologists, highlighting the critical need for automated, objective analytical tools to augment clinical decision-making⁹.

Recent advances in Artificial Intelligence, particularly deep learning, present a powerful opportunity to automate the objective quantification of AS-OCT images^{10,11}. Deep learning methodologies now effectively extract

¹School of Future Technology, South China University of Technology, Guangzhou, China. ²HDMI Lab, South China University of Technology, Guangzhou, China. ³School of Biomedical Sciences and Engineering, South China University of Technology, Guangzhou, China. ⁴Shenzhen Eye Hospital, Shenzhen Eye Medical Center, Southern Medical University, Shenzhen, China. ⁵These authors contributed equally: Jiongning Zhao, Huihui Fang. ✉e-mail: chiwei@mail.sysu.edu.cn; benben0606@139.com

Dataset	Year	Resolution	Picture	Patient	Labels		Categories			
					Localization	Mask	Normal	Glaucoma	Cataract	Glaucoma & Cataract
AGE ⁴²	2019	998 × 998; 224 × 224	4800	199	✓	×	✓	✓	×	×
AIDK ⁴³	2024	1657 × 1000; 1723 × 1000	FFD:768; PFD:400	64	×	✓	×	×	×	×
MCOA ^{44–46}	2025	3900 × 2500	6272	300	×	✓	✓	✓	×	×
MSDC	2025	2035 × 1868	1106	627	✓	✓	✓	✓	✓	✓

Table 1. Summary of publicly available AS-OCT datasets. *Note:* MSDC: ASOCT-MSDC dataset proposed in this work.

quantitative biometrics from AS-OCT scans. Fully convolutional networks achieve precise delineation of micro-structural boundaries, such as rapid pachymetric mapping via CorneaNet¹² and dual-stream edge detection via BG-CNN¹³ to mitigate boundary smoothing. Multi-task models like ScLNet¹⁴ extend these capabilities to prosthetic interfaces, accurately segmenting scleral lenses and tear fluid reservoirs. For 3D volumetric consistency across B-scans, sequence correction networks like SeqCorr-EUNet¹⁵ resolve spatial discontinuities. Concurrently, automated scleral spur localization has advanced through probability heatmap regression and L1-optimized coordinate regression¹⁶, demonstrating robust clinical viability with minimal median errors on commercial platforms.

Beyond structural parsing, end-to-end architectures directly classify anterior segment pathologies from pixel intensities. In primary angle-closure disease (PACD) screening, classifiers achieve high accuracy in detecting occludable angles¹⁷, while spatiotemporal sequence analysis using ConvLSTM-TC networks captures dynamic iris deformations to improve specificity¹⁸. Furthermore, models can infer hidden sub-surface conformations, such as plateau iris syndrome, directly from surface topographies¹⁹. In cataract management, deep feature extractors objectively quantify lenticular opacities to automate subjective grading systems like LOCS III²⁰.

Despite these milestones, clinical translation remains hindered by data-centric bottlenecks. Models trained on homogeneous data suffer severe domain shifts across different OCT hardware, requiring cross-domain translation frameworks like GANSeg to synthesize domain-agnostic features^{21,22}. Existing algorithms are also highly sensitive to physiological artifacts²³; for instance, networks may paradoxically anchor predictions on eyelid obscuration or reflections rather than actual pathology²⁴. Additionally, standard softmax classifiers assume isolated pathologies, rendering them ill-equipped for complex, co-occurring diseases common in geriatric populations and necessitating multi-task architectures²⁵. These vulnerabilities underscore a profound dataset scarcity, forcing researchers to increasingly rely on synthetic data generation via GANs to artificially augment training distributions²⁶.

However, the development of clinically robust AI models is fundamentally constrained by the availability of suitable training data. As shown in Table 1, a survey of existing public datasets reveals several critical shortcomings that hinder progress. First, the overall scarcity of large-scale, high-quality datasets with consistent, expert-level annotations—as well as explicitly documented image quality information—remains a major barrier to advancing the field. Second, many available datasets are limited in scope, focusing on a single pathology, such as keratitis, or a narrow set of anatomical structures, often omitting the detailed intra-lenticular segmentation vital for quantitative cataract analysis²⁷. Finally, a common methodological flaw in data set design is the inclusion of multiple images from the same patient. This practice can lead to data leakage between training and test sets, creating a significant risk of developing models with overly optimistic performance metrics that fail to generalize to new unseen individuals²⁸.

To address these specific limitations, we present the ASOCT-MSDC (AS-OCT Multi-Structure and Disease Classification Dataset), a comprehensively annotated resource designed to facilitate the development and evaluation of deep learning algorithms in ophthalmic imaging. The dataset contains 1,106 AS-OCT images from 1,106 eyes of 627 individuals across four clinical categories: normal, cortical cataract, PACG, and PACG with cortical cataract, with representative examples shown in Fig. 1. The distribution of images by clinical category and eye laterality is summarized in Table 2. Each image is accompanied by anatomical segmentation masks, scleral-spur landmarks, diagnostic labels, and image-quality scores. To reduce subject-level leakage, we ensured that each eye contributes at most one high-quality scan. This design supports the development and evaluation of generalizable models for anatomical segmentation, scleral-spur localization, and disease classification in anterior segment analysis.

Our key contributions of this work are threefold:

1. We present a curated dataset of 1,106 AS-OCT images from 1,106 eyes of 627 individuals across four clinical categories: normal, cortical cataract, PACG, and PACG with cortical cataract, providing a reliable resource for developing and evaluating generalizable learning algorithms.
2. The dataset provides expert-validated annotations of key anterior segment structures and scleral spur locations. The annotations were produced by trained technicians and verified by an experienced anterior segment specialist, ensuring high-quality and consistent structural labeling.
3. To demonstrate the usability of the dataset, we provide baseline experimental results for multiple tasks, including anatomical structure segmentation, scleral spur localization, and disease classification, using representative deep learning models.

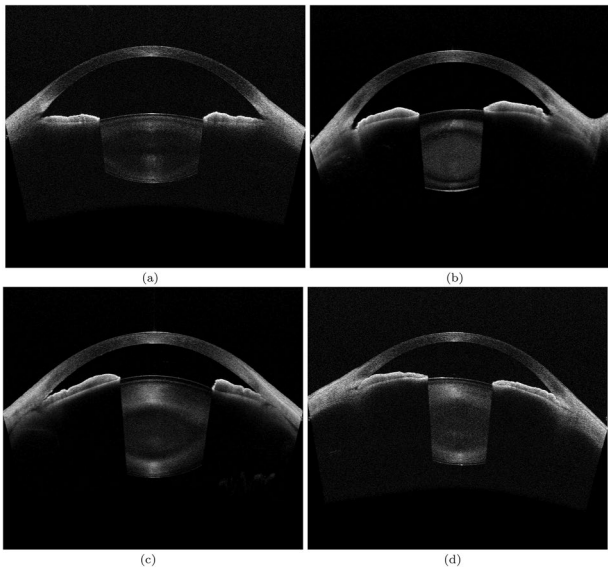


Fig. 1 Representative AS-OCT images from the four clinical categories in ASOCT-MSDC. **(a)** Normal, **(b)** cataract, **(c)** glaucoma, and **(d)** glaucoma with cataract. In this dataset, the cataract category refers to cortical cataract, and the glaucoma category refers to PACG.

Cases	Number of Images	Eye Distribution	Number of Images
Normal	326 (29.5 %)	OD	164 (50.3 %)
		OS	162 (49.7 %)
Cataract	499 (45.1 %)	OD	243 (48.7 %)
		OS	256 (51.1 %)
Glaucoma	143 (12.9 %)	OD	70 (49.0 %)
		OS	73 (51.1 %)
Glaucoma & Cataract	138 (12.5 %)	OD	65 (47.1 %)
		OS	73 (52.9 %)

Table 2. Population Distribution.

Methods

The entire working flow of this dataset establishment can be shown in Fig. 2.

Ethics statement. The study was approved by the Institutional Review Board and Ethics Committee of Shenzhen Eye Hospital (Approval No. [2025KYPJ204]), and all procedures adhered to the ethical principles outlined in the Declaration of Helsinki. All imaging data and accompanying metadata were fully de-identified prior to analysis, with personal identifiers (such as name and medical record number) removed at the source system. Only non-identifiable fields relevant to scientific use—disease category, eye laterality, acquisition date, and three quality assessment scores—were retained. Study-specific serial numbers were generated programmatically to maintain traceability while ensuring complete anonymity. All data were retrospectively collected from clinical records. The requirement for informed consent was waived by the Ethics Committee due to the use of fully anonymized historical clinical data and the minimal-risk nature of the study.

Patient Selection. Clinical AS-OCT images were collected from patients examined at the Ophthalmology Department of Shenzhen Eye Hospital between February 2022 and December 2025. Eligible images were required to have been acquired using the standardized CASIA2 AS-OCT protocol and to meet the predefined image-quality criteria. Disease groups were retrospectively identified from routine clinical records, whereas normal controls were selected from ophthalmology outpatient individuals with no clinically detectable ocular disease.

All glaucoma and glaucoma-cataract cases in ASOCT-MSDC were clinically diagnosed as primary angle-closure glaucoma (PACG). No open-angle glaucoma cases were included. In this manuscript, the glaucoma category therefore refers to PACG, and the cataract category refers to cortical cataract unless otherwise specified. PACG diagnosis followed the Expert Consensus on the Diagnosis and Treatment of Primary Angle-Closure Glaucoma in China (2019) and the Chinese Glaucoma Guidelines (2020)^{29,30}. Gonioscopy was used as the reference standard for angle assessment. Patients also underwent intraocular pressure measurement, anterior segment examination, and fundus OCT examination; visual-field testing was performed in a subset of patients when clinically indicated for diagnostic support and disease-severity assessment.

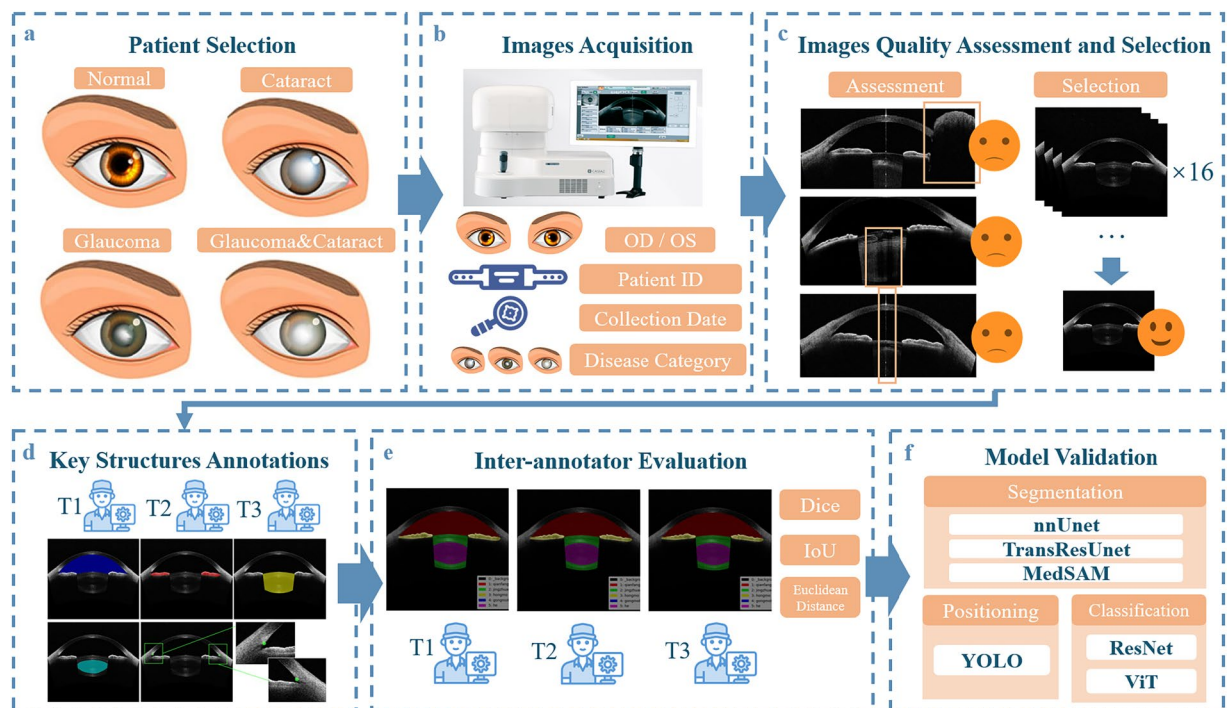


Fig. 2 Flow chart of the ASOCT-MSDC dataset. **(a)** The study included 1,106 eyes from patients with cataract, glaucoma, coexisting glaucoma and cataract, and normal controls. **(b)** AS-OCT images were captured using Tomy CASIA2; recorded data included eye laterality (OD/OS), acquisition date, and disease category. **(c)** Technicians assessed and selected images based on three criteria: eyelid obscuration, black-line anomaly, and central light artifact, with expert review. **(d)** Seven key structures were annotated: anterior chamber, iris, lens, lens nucleus, anterior cortex, posterior cortex, and scleral spurs (SS). **(e)** Three technicians independently annotated 20 randomly selected images, and inter-rater reliability was improved through training. **(f)** We validated structure segmentation using nnU-Net, TransResU-Net, and MedSAM, used YOLOv11l-pose for scleral-spur localization, and used ResNet50 and ViT-Small for disease classification.

Normal controls were selected from individuals examined in the ophthalmology outpatient clinic. They had no anterior segment disease, no posterior segment disease, no cataract, no glaucoma, and no history of ocular surgery. Normal ocular status was confirmed by comprehensive ophthalmic examination, including slit-lamp examination, intraocular pressure measurement, and fundus photography.

Cataract cases in ASOCT-MSDC were clinically diagnosed as cortical cataract by experienced ophthalmologists based on slit-lamp examination. The glaucoma-cataract category consisted of eyes with coexisting PACG and cortical cataract. Standardized LOCS III grading records, original slit-lamp photographs, and demographic variables were not retained in the final released research database after institutional de-identification and data-minimization procedures during data export. Therefore, ASOCT-MSDC does not provide LOCS-based cataract severity labels. The current cataract category represents cortical cataract and should not be interpreted as covering all cataract subtypes or supporting LOCS-based cataract severity grading.

Image acquisition. All AS-OCT images were acquired using a Swept-Source OCT device, the Tomy CASIA2. This device operates at a wavelength of 1310 nm, with a scanning speed of 50,000 scans per second, providing an axial resolution of $<10\ \mu\text{m}$ in tissue. The acquisitions were performed by trained ophthalmic technicians following a standardized protocol. Each eye was scanned using the “Radial” scan mode, which captures 16 cross-sectional B-scans centered on the corneal apex. During the examination, patients were instructed to fixate on an internal target to minimize eye movement. Eye laterality (OD: oculus dexter; OS: oculus sinister), and collection date were recorded at the same time. The scoring criteria for eyelid obscuration, black-line anomaly, and central light artifact are illustrated in Figs. 3, 4, and 5, respectively.

Image Quality Assessment and Selection. Image quality assessment and selection were performed by three trained technicians who were not involved in image acquisition. The procedure consisted of two sequential stages. In the first stage, all available radial B-scans for each eye were inspected, and one representative B-scan was selected according to a predefined best-view selection protocol. Image quality was assessed using three criteria: eyelid obscuration, black-line anomaly, and central light artifact. Each criterion was scored as 0, 1, or 2, corresponding to optimal quality, suboptimal but usable quality, and unacceptable quality, respectively. When multiple B-scans were available for the same eye, best-view selection followed the predefined priority order of eyelid obscuration, black-line anomaly, and central light artifact.

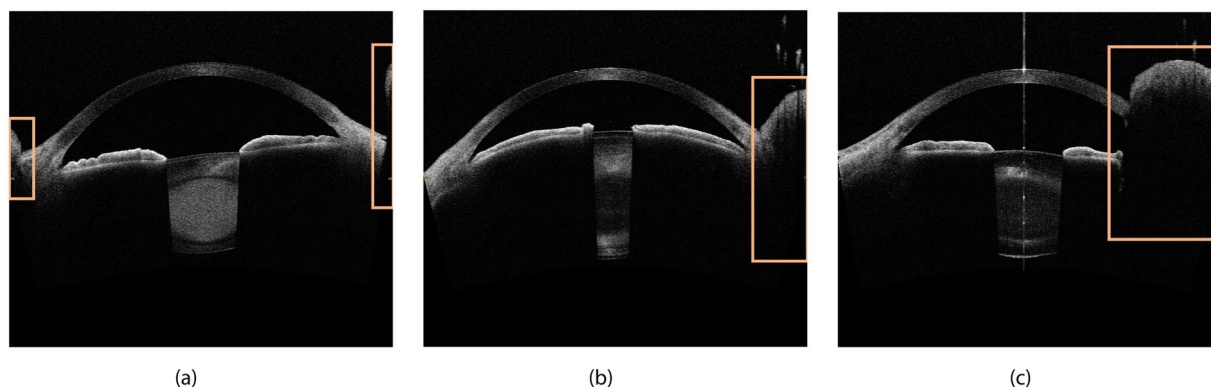


Fig. 3 Eyelid Obscuration Assessment Criteria. (The eyelids are marked with boxes in the picture.) (a) 0: No or minimal obscuration by the eyelids, and no interference with the assessment of critical structures. (b) 1: Partial obscuration by the eyelids is present, but the scleral spur is visible. (c) 2: Severe obscuration by the eyelids is present, and the scleral spur is not visible.

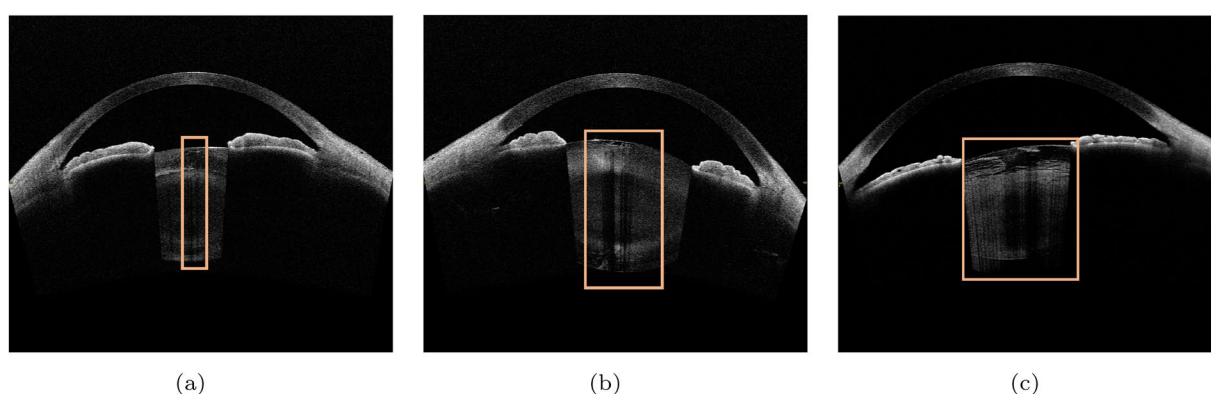


Fig. 4 Black Lines Anomaly Assessment Criteria. (The black lines are marked with boxes in the picture.) (a) 0: No or minimal presence of edema artifact or eyelash shadowing, allowing for the assessment of critical structures. (b) 1: Significant edema artifact is present, causing the black areas within the structure or poorly demarcated boundaries. (c) 2: Severe edema artifact is present, causing the black areas to account for most of the structure's interior or unrecognizable boundaries.

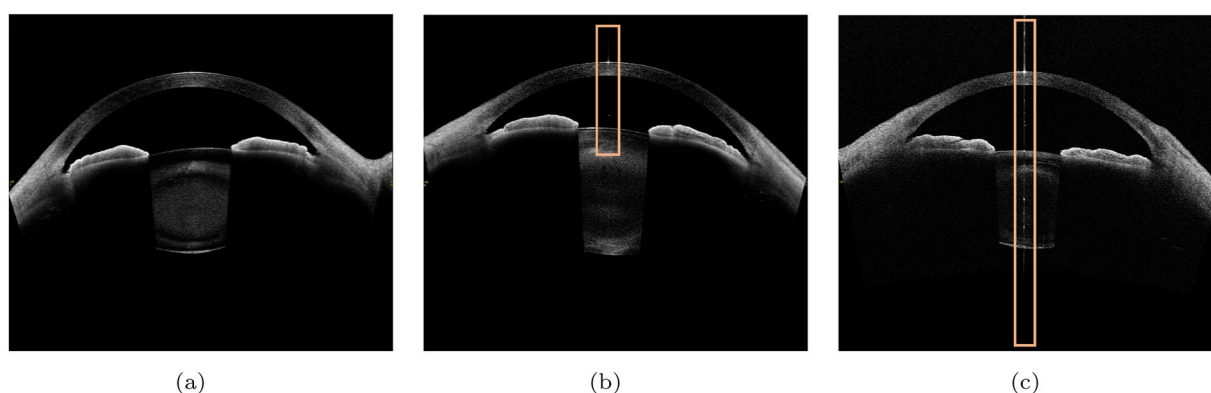


Fig. 5 Central Light Artifact Assessment Criteria. (The central light artifacts are marked with boxes in the picture.) (a) 0: No central light artifact visualized. (b) 1: Fine, discontinuous central light artifact is present. (c) 2: Thick, continuous central light artifact that traverses the image is present.

In the second stage, the preselected representative B-scans underwent strict quality exclusion. Representative B-scans with a score of 2 for any of the three criteria were excluded from the final dataset. Uncertain cases were adjudicated by a physician with more than 10 years of experience.

Category	Representative B-scans after Stage 1	Excluded after Stage 2	Eyelid obscuration	Black-line anomaly	Central light artifact	Final included images
Normal	328	2	0	0	2	326
Cataract	523	24	6	15	3	499
Glaucoma	146	3	0	3	0	143
Glaucoma & Cataract	142	4	3	1	0	138
Total	1,139	33	9	19	5	1,106

Table 3. Two-stage image-quality screening and inclusion summary. Stage 1 refers to best-view preselection, in which one representative B-scan was selected for each eye from the available radial B-scans. Stage 2 refers to strict quality exclusion, in which preselected representative B-scans with a score of 2 for any quality criterion were excluded. Exclusion reasons are reported according to the primary unacceptable quality criterion. In ASOCT-MSDC, the cataract category refers to cortical cataract, and the glaucoma category refers to PACG.

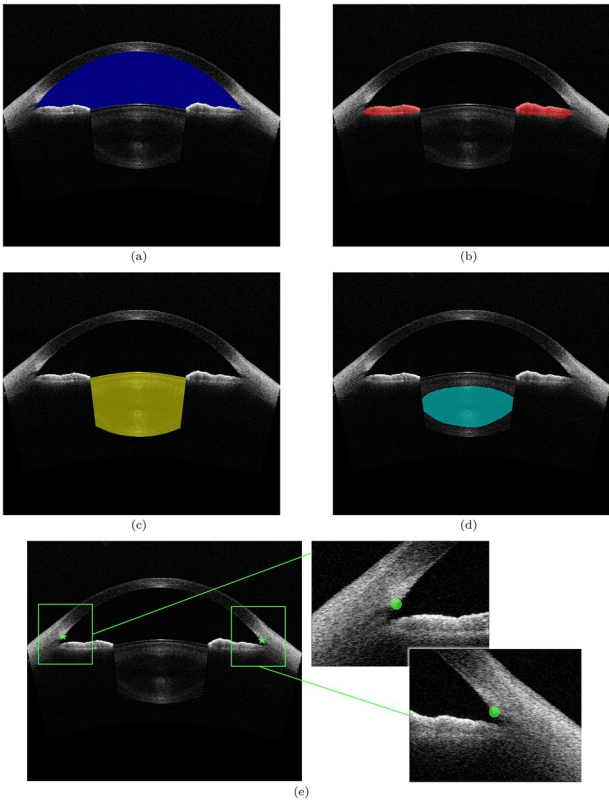


Fig. 6 The annotated diagram in the AS-OCT images. (a) Anterior chamber (b) Iris (c) Lens (d) Nucleus (e) Scleral spur.

The initial raw image pool contained more than 18,000 radial B-scans, with approximately 16 B-scans typically available per eye. During the first-stage best-view preselection, 1,139 representative B-scans were selected, one for each eye. During the second-stage strict quality exclusion, 33 representative B-scans were excluded because at least one quality criterion received a score of 2. The primary exclusion reasons were severe eyelid obscuration in 9 B-scans, severe black-line anomaly in 19 B-scans, and severe central light artifact in 5 B-scans. After the two-stage selection and exclusion process, 1,106 images were included in the final dataset, as summarized in Table 3.

Key Structure Annotations. The core annotation work was performed by a professional team of three technicians who had received standardized, intensive training. The annotation tool is Labelme³¹. The training was led by a senior ophthalmologist with over 10 years of clinical experience in anterior segment diseases, focusing on precise identification of anatomical structures according to established ophthalmology definitions.

As is shown in Fig. 6, for each image, seven labels were generated, including polygon masks for segmentation and point-by-point markers. The polygon masks included the Anterior Chamber (AC), Iris, Lens, and Lens Nucleus. The Anterior Chamber (AC) is defined as the area bounded by the posterior surface of the cornea, the anterior surface of the iris, and the anterior capsule of the lens. The Iris is the tissue from the root to the pupillary margin. The Lens is the area bounded by its anterior and posterior capsules. The Lens Nucleus, Anterior Cortex,

First-level	Second-level	Third-level	Fourth-level
Images	Normal Folder	Images Folder	Images (JPG, n=326)
		Annotated Folder	Annotations (JSON, n=326)
	Cataract Folder	Images Folder	Images (JPG, n=499)
		Annotated Folder	Annotations (JSON, n=499)
	Glaucoma Folder	Images Folder	Images (JPG, n=143)
		Annotated Folder	Annotations (JSON, n=143)
	Glaucoma_Cataract Folder	Images Folder	Images (JPG, n=138)
		Annotated Folder	Annotations (JSON, n=138)
Texts	Metadata_Normal (XLSX)		
	Metadata_Cataract (XLSX)		
	Metadata_Glaucoma (XLSX)		
	Metadata_Glaucoma_Cataract (XLSX)		

Table 4. Data Structure and Distribution.

and Posterior Cortex were segmented as intra-lenticular structures based on visible changes in signal intensity and texture on the ASOCT images, corresponding to their anatomical boundaries. The point-by-point marker is the Scleral Spur (SS), which is a pair of points marking the innermost protrusion of the sclera at the posterior end of the trabecular meshwork.

Based on established ophthalmology definitions, three technicians independently annotated a random selection of 20 images. The senior ophthalmologist reviewed these initial annotations, provided detailed feedback, and moderated discussions to resolve any discrepancies. This iterative process was repeated until a high level of consensus was achieved across all annotation criteria.

After confirming the high inter-observer consistency among the three technicians, the complete main dataset was assigned to them for formal annotation. Technician No. 1 annotated images No. 0-51 of the normal cases and images No. 0-229 of the cataract cases. Technician No. 2 annotated images No. 230-508 of the cataract cases and images No. 0-2 of the glaucoma cases. Technician No. 3 annotated images No. 3-146 of the glaucoma cases and images No. 0-140 of the glaucoma&cataract cases. These annotations provide critical input for model learning, aiding in the segmentation of anterior segment structures, the detection of the scleral spur’s position, and ultimately assisting in the diagnosis of eye diseases.

Quality Control and Validation. After completion of the initial annotation, all 1,106 annotated images were reviewed by a senior anterior segment ophthalmologist. Annotations that were inconsistent with the pre-defined anatomical protocol were returned to the corresponding technician for correction and resubmission. In total, 129 images (11.66%) required at least one correction. The main correction types included scleral-spur localization adjustment, lens-boundary refinement, and refinement of weak or ambiguous anatomical boundaries. All corrected annotations were re-submitted and re-evaluated by the senior ophthalmologist. The review-correction-re-review cycle was repeated until all annotations were accepted.

Data Records

The ASOCT-MSDC dataset is available at the public figshare repository³². To ensure systematic handling of the imaging data, a hierarchical directory structure was designed to separate cases according to clinical categories and eye laterality, as shown in Table 4. The dataset was organized under four major categories: normal, cataract, glaucoma, and glaucoma&cataract. Each category folder contains two subfolders, in which two types of files are stored respectively:

- 1. Raw images in “jpg” format, representing the anterior segment optical coherence tomography (AS-OCT) scan of the eye.
- 2. Annotation files in “json” format, containing the corresponding labeling and metadata necessary for supervised analysis.

1106 raw images were numbered one by one, and the naming rule was “n.jpg”, where n is a digit. Apart from the extension, each original image and its corresponding annotation file have the same file name. Cases of normal, cataract, glaucoma, and glaucoma&cataract follow an identical organizational scheme to ensure consistency across different disease groups. This arrangement facilitates easy navigation and scalable extension of the dataset.

In addition, metadata texts are maintained to record eye, acquisition time and image quality associated with each image. These files provide a structured reference that links imaging data with clinical attributes, enabling comprehensive analysis of disease-related features.

This directory design not only standardizes data storage but also ensures traceability, reproducibility, and interoperability for subsequent machine learning tasks, such as model training, validation, and external evaluation. The separation of raw images, annotations, and metadata allows for flexible preprocessing and integration into diverse computational pipelines.

	Anterior Chamber		Lens		Nucleus		Iris		SS Pixel Distance
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU	
Before training									
Normal	0.9882	0.9768	0.8549	0.8014	0.8053	0.7537	0.9033	0.8247	16.76
Cataract	0.9804	0.9617	0.9392	0.8859	0.9615	0.9253	0.8833	0.7928	24.59
Glaucoma	0.9807	0.9623	0.9024	0.8237	0.8750	0.7840	0.9055	0.8281	24.53
Glaucoma& Cataract	0.9826	0.9659	0.9383	0.8840	0.9627	0.9287	0.9242	0.8594	34.21
After training									
Normal	0.9921	0.9843	0.9907	0.9815	0.9557	0.9157	0.9577	0.9190	7.601
Cataract	0.9883	0.9780	0.9887	0.9777	0.9641	0.9310	0.9538	0.9118	7.893
Glaucoma	0.9825	0.9751	0.9859	0.9729	0.9676	0.9374	0.9566	0.9167	10.76
Glaucoma & Cataract	0.9826	0.9659	0.9863	0.9730	0.9627	0.9287	0.9604	0.9240	11.47

Table 5. Performance metrics across different conditions and anatomical structures.

Technical Validation

Inter-annotator Validation. To rigorously assess the consistency and reliability of the annotation protocol, an inter-annotator validation was conducted on a dedicated subset of 20 randomly selected AS-OCT images stratified across the four clinical categories. Crucially, during this validation phase, all three trained technicians independently annotated the identical set of 20 images. This shared subset was specifically designed to quantify inter-rater agreement and establish a reference standard for human annotation consistency. The inter-annotator agreement on this shared subset was quantitatively evaluated using three metrics: Euclidean distance for scleral spur localization, and Intersection-over-Union (IoU) and Dice coefficient for structural segmentation. The Euclidean distance measures the pixel-level positional discrepancy between corresponding landmark annotations, reflecting the spatial consistency of scleral spur localization across annotators. For ground-truth coordinates (x, y) and reference coordinates (x, y) , it is defined as:

$$d = \sqrt{(x - x)^2 + (y - y)^2}. \quad (1)$$

IoU and Dice evaluate segmentation accuracy by calculating the spatial overlap between the model's predicted mask A and the manually annotated ground-truth mask $B^{33,34}$. IoU is defined as the ratio of the intersection area to the union area, while the Dice coefficient is defined as twice the intersection area divided by the sum of both areas:

$$\text{IoU} = \frac{|A \cap B|}{|A \cup B|}, \quad (2)$$

$$\text{Dice} = \frac{2 \times |A \cap B|}{|A| + |B|}. \quad (3)$$

The average coefficients for inter-annotator validation before and after training are summarized in Table 5. This table demonstrates a marked improvement in consistency following the training phase. Eventually, all structural segmentation coefficients (Dice and IoU) surpassed 90%, indicating a high degree of reliability among the three technicians. To provide a granular analysis of pairwise consistency, the results are further visualized in Fig. 7. Figure 7(a) illustrates the landmark localization consistency by plotting the boxplots of pixel-level Euclidean distances for the left and right Scleral Spurs between each pair of annotators (E1 vs. E2, E1 vs. E3, and E2 vs. E3). While a slightly larger variance was observed in glaucoma and co-existing cases due to anatomical angle closure, the overall localization error remained minimal. Figure 7(b) and (c) detail the segmentation agreement for specific anatomical structures. These charts display the Intersection-over-Union (IoU) and Dice coefficients, respectively, for the Anterior Chamber, Lens, Nucleus, and Iris. The uniformly high scores across all pairwise comparisons confirm that the annotators achieved a stable consensus on anatomical boundaries.

To rigorously validate the reliability of the manual annotations, we assessed the agreement among the three independent technicians using the Dice and IoU on a validation subset. We performed a One-way Analysis of Variance (ANOVA) followed by pairwise comparisons on the score distributions to statistically evaluate the consistency. As visualized in Fig. 8, the analysis revealed no statistically significant differences in annotation performance among the three groups. For DSC, the ANOVA yielded a P-value of 0.4700 ($P > 0.05$) and for IoU, the P-value was 0.4561 ($P > 0.05$). Additionally, all pairwise comparisons were non-significant. These results, combined with high mean scores, statistically confirm that the annotations are robust, unbiased, and consistent across different technicians.

Segmentation Validation. To demonstrate the practical utility of the constructed dataset and provide baseline metrics for future comparisons, we implemented a series of semantic segmentation experiments. Three representative deep learning models were selected for evaluation: nnU-Net³⁵, TransResU-Net³⁶, and MedSAM³⁷. These three models were selected because they represent complementary segmentation paradigms.

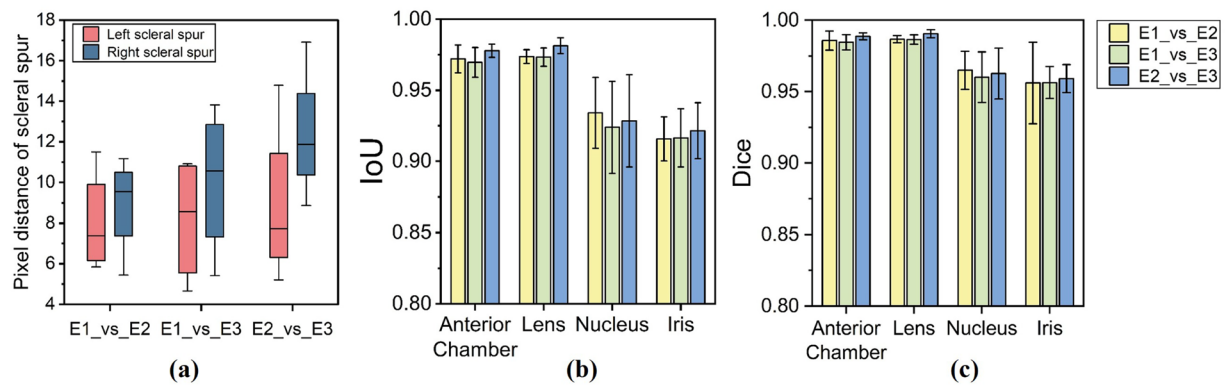


Fig. 7 Annotation results. **(a)** Evaluation results of average pixel distance between each annotating scleral spur on the same series of images by different annotators. **(b)** **(c)** Evaluation results of IoU and Dice between each annotations on the same series of images by different annotators.

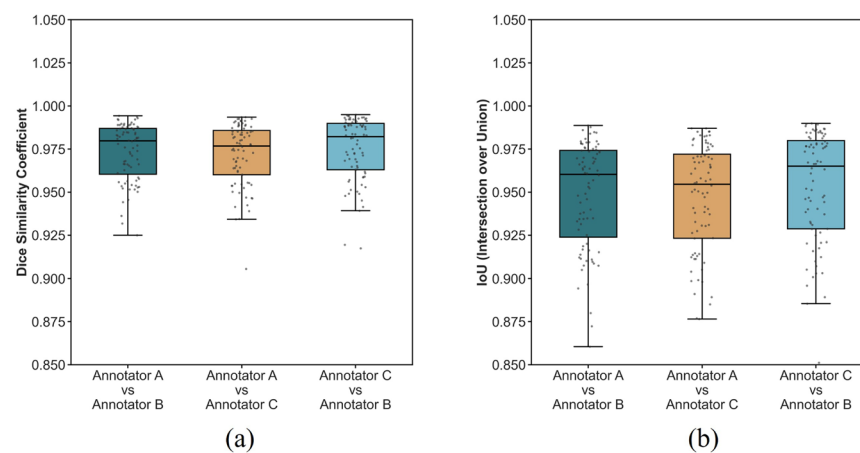


Fig. 8 Inter-annotator consistency analysis of Dice and IoU similarity coefficients.

nnU-Net was included as a strong self-configuring medical image segmentation baseline that automatically adapts preprocessing, network configuration, and training settings to the target dataset. TransResU-Net was selected as a CNN-Transformer hybrid architecture that combines convolutional local feature extraction with transformer-based global context modeling. MedSAM was included as a medical foundation-model-based segmentation approach, allowing us to evaluate how a prompt-driven model pretrained on large-scale medical images transfers to AS-OCT anatomy. Together, these baselines provide a practical comparison across conventional medical segmentation, hybrid CNN-Transformer modeling, and foundation-model-based segmentation.

The dataset was divided into training, validation, and test sets at a ratio of 70%, 15%, and 15%, respectively, using a stratified strategy based on diagnostic categories. The partition was performed at the participant level to avoid data leakage across subsets. The training set was used for gradient-based parameter optimization. The validation set was used only for hyperparameter tuning, early stopping, and checkpoint selection and was not involved in gradient-based optimization. The test set was kept fixed and was used only for final performance reporting. The same predefined partitioning strategy was used for the baseline experiments to support fair comparison.

For nnU-Net and TransResU-Net, we trained independent models for four anatomical structures: anterior chamber, lens, iris, and nucleus. When training a model for a specific structure, the corresponding mask region was treated as having a pixel value of 1, while all other regions were treated as having a pixel value of 0. This process was repeated sequentially for the four target regions. Consequently, both nnU-Net and TransResU-Net generated four independent sets of optimal model parameters tailored to specific anatomical structures. For MedSAM, we adopted a fine-tuning strategy from transfer learning. We first loaded the officially released pre-trained weights, which had been trained on a large-scale dataset of medical images. The prompt encoder was frozen, whereas both the image encoder and mask decoder were optimized using the AS-OCT training images.

Finally, the IoU and Dice scores for all samples on the independent test set for each model and anatomical structure are computed, taking their arithmetic mean as the final performance metric. Detailed evaluation results are shown in the Table 6.

The segmentation results demonstrated clear structure-dependent differences across anatomical regions and models. The anterior chamber achieved consistently high segmentation performance, likely because it forms a

Anatomical Structure	Method	Dice (%)	IoU (%)
Anterior Chamber	nnU-Net	0.9878 ± 0.0067	0.9760 ± 0.0126
	TransResU-Net	0.9903 ± 0.0031	0.9808 ± 0.0061
	MedSAM	0.9788 ± 0.0294	0.9598 ± 0.0462
Lens	nnU-Net	0.9454 ± 0.0071	0.9038 ± 0.0088
	TransResU-Net	0.9889 ± 0.0245	0.9790 ± 0.0369
	MedSAM	0.7536 ± 0.3052	0.6783 ± 0.3032
Nucleus	nnU-Net	0.9238 ± 0.0099	0.8889 ± 0.0200
	TransResU-Net	0.9270 ± 0.0882	0.9270 ± 0.0810
	MedSAM	0.9412 ± 0.0336	0.8907 ± 0.0567
Left Iris	nnU-Net	0.7623 ± 0.0935	0.6914 ± 0.0952
	TransResU-Net	0.8458 ± 0.1307	0.7501 ± 0.1554
	MedSAM	0.9213 ± 0.0262	0.8552 ± 0.0437
Right Iris	nnU-Net	0.8517 ± 0.0230	0.7657 ± 0.0345
	TransResU-Net	0.8617 ± 0.1150	0.7689 ± 0.1195
	MedSAM	0.9157 ± 0.0253	0.8455 ± 0.0420

Table 6. Segmentation Results.

relatively large, homogeneous, and high-contrast region with continuous anatomical boundaries. Lens segmentation also showed high Dice and IoU scores for the fully supervised nnU-Net and TransResU-Net models, with TransResU-Net achieving the best performance. This result suggests that supervised models trained directly on AS-OCT annotations can effectively learn the global lens shape and preserve its overall contour.

MedSAM showed substantially lower performance for whole-lens segmentation (Dice, 0.7536) than nnU-Net (0.9454) and TransResU-Net (0.9889), despite performing well on the smaller lens nucleus and iris targets, indicating that anatomical size alone does not explain the discrepancy. Whole-lens segmentation requires recovery of the posterior lens capsule, which lies in a deeper region and may be weak or discontinuous because of OCT signal attenuation and tissue scattering. In addition, the whole-lens bounding box contains nested nucleus and cortex structures and heterogeneous cataract-related signals. Because a box prompt conveys spatial extent but not anatomical hierarchy, MedSAM may favor salient internal regions and produce incomplete or fragmented outer-lens masks, as illustrated in Fig. 9.

By contrast, the nucleus is compact and centrally localized, while the left and right iris boxes impose stronger spatial constraints, reducing prompt ambiguity. The large standard deviation of the MedSAM lens Dice score suggests that the lower mean was driven mainly by unstable predictions in a subset of images rather than small, uniform errors across the test set. These findings indicate that prompt-driven models may require AS-OCT-specific adaptation, such as multi-scale feature fine-tuning, boundary-aware supervision, shape or connectivity constraints, and prompts that better distinguish the outer capsule from nested lens structures.

Nucleus segmentation showed intermediate-to-high performance, but errors were more likely to occur when the nucleus-cortex interface was faint, discontinuous, or affected by cortical cataract-related signal heterogeneity. This suggests that segmentation of internal lens substructures is more dependent on subtle texture and boundary contrast than segmentation of larger external anatomical regions.

Iris segmentation was comparatively more challenging than anterior chamber and lens segmentation because the iris is thinner, locally low-contrast, and may be partially obscured near the iridocorneal angle. In addition, separating the left and right iris as two independent anatomical labels further increases the difficulty, as models may occasionally confuse the two sides or produce incomplete segmentation in regions with angle crowding. The representative right iris example in Fig. 9 shows that the fully supervised models generally preserved a more anatomically plausible iris contour, whereas MedSAM showed less stable delineation in the thin peripheral iris region. These visual examples support the quantitative findings and further demonstrate that segmentation performance in AS-OCT is strongly influenced by anatomical size, boundary contrast, tissue heterogeneity, and model-specific adaptation to the imaging modality.

Positioning Validation. We evaluated the feasibility and reliability of scleral spur localization in AS-OCT images using a keypoint detection framework based on YOLOv11¹¹-pose. The model was trained in a fully supervised manner with manually annotated left and right scleral spur coordinates and directly regressed the (x, y) locations of bilateral scleral spur landmarks during inference. Performance was evaluated from two complementary perspectives: (i) pose-level detection metrics, including precision (P), recall (R), and mean average precision (mAP), and (ii) pixel-level localization accuracy measured by the mean pixel distance (MPD) and the percentage of correct keypoints (PCK) under multiple pixel thresholds³⁸.

The mean pixel distance (MPD) is computed as

$$\text{MPD} = \frac{1}{N} \sum_{i=1}^N d_i, \quad (4)$$

and the percentage of correct keypoints (PCK) at a pixel threshold k is defined as

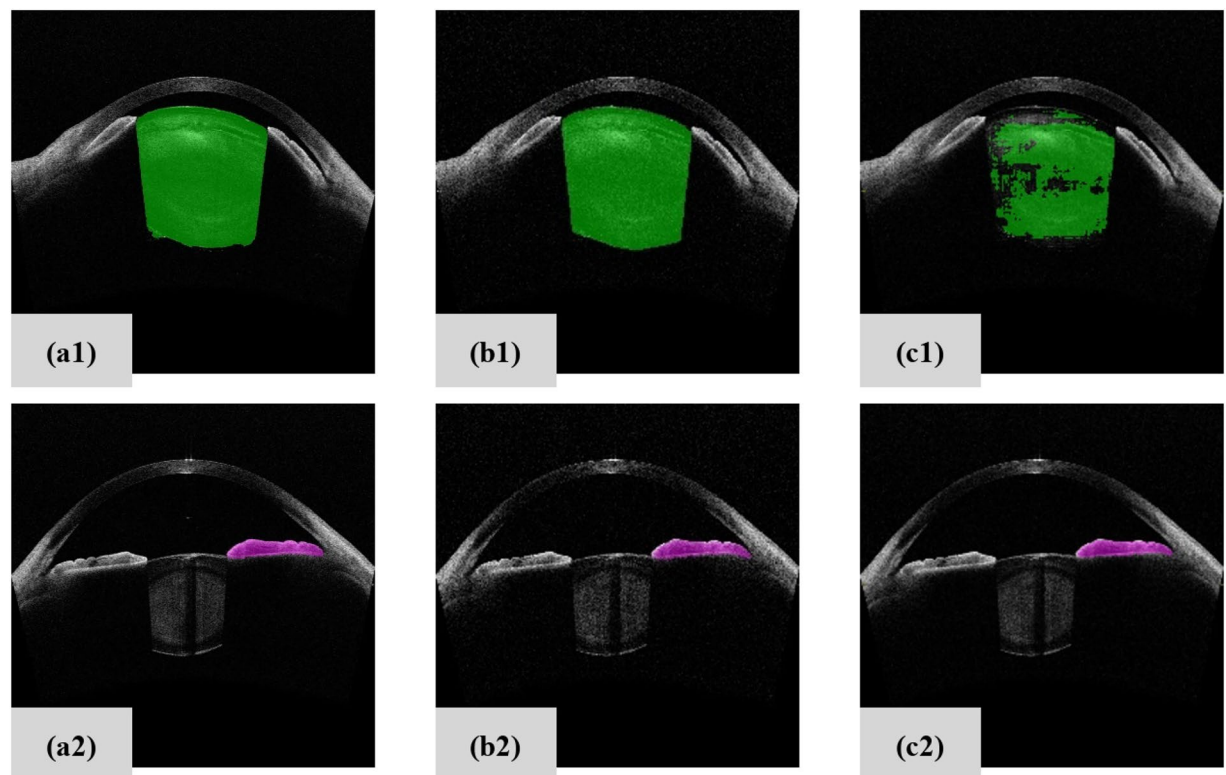


Fig. 9 Representative segmentation results for the lens and right iris on the independent test set. The upper row shows lens segmentation results from (a1) nnU-Net, (b1) TransResU-Net, and (c1) MedSAM. The lower row shows right-iris segmentation results from (a2) nnU-Net, (b2) TransResU-Net, and (c2) MedSAM. In the lens example, MedSAM produced an incomplete, fragmented mask near the weak posterior lens boundary. The right-iris example illustrates the challenge of segmenting thin and low-contrast iris tissue near the angle region. These representative cases visually demonstrate the structure-dependent performance differences observed in the quantitative segmentation results.

$$\text{PCK}@k = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(d_i < k), \quad (5)$$

where $\mathbf{1}(\cdot)$ denotes the indicator function.

As summarized in Table 7, the proposed model achieved an overall MPD of 22.45 pixels with a median distance of 19.72 pixels. The corresponding PCK values reached 18.1%, 51.4%, and 95.2% at thresholds of 10, 20, and 50 pixels, respectively, indicating that most predicted scleral spur locations were constrained within anatomically plausible regions, while precise boundary localization remained challenging at stricter thresholds. At the pose level, the overall detection performance was stable, with a precision of 0.868, recall of 0.860, and mAP of 0.891.

A category-wise analysis revealed clear performance differences across clinical conditions. The normal cohort achieved the best localization accuracy, with the lowest MPD (18.41 pixels) and the highest PCK values (61.7% at 20px and 100.0% at 50px), accompanied by excellent pose-level detection performance (mAP = 0.995). In contrast, glaucoma-related cases exhibited consistently inferior localization accuracy. The glaucoma cohort showed reduced pose-level performance (mAP = 0.774) and lower pixel-level accuracy (MPD = 24.60 pixels; PCK@20px = 53.2%), while the glaucoma&cataract cohort presented the largest localization error (MPD = 28.58 pixels) and the lowest PCK@20px (34.2%). These results indicate that the coexistence of multiple pathological factors further exacerbates the difficulty of precise scleral spur localization.

In advanced PACG, the scleral spur may be genuinely unresolvable on a single B-scan; forcing every automated prediction to return a coordinate can therefore create false precision. A clinically oriented system should combine localization with visibility assessment or calibrated confidence, abstain from low-confidence predictions, and route such scans for expert review or assessment using adjacent B-scans or volumetric context. Future training sets could additionally distinguish visible, ambiguous, and non-visible landmarks and retain multi-rater uncertainty for difficult cases. The current dataset provides expert-adjudicated reference coordinates but no separate landmark-visibility or uncertainty label, so the present benchmark should not be interpreted as validating fully autonomous localization when the scleral spur is not discernible.

Category	Detection Metrics			Error (px)		PCK (Accuracy)		
	P	R	mAP95	MPD	Median	10px	20px	50px
Normal	0.953	1.000	0.995	18.41	16.86	22.3%	61.7%	100.0%
Cataract	0.965	0.934	0.987	22.74	20.40	19.1%	48.7%	94.7%
Glaucoma	0.700	0.875	0.774	24.60	19.68	12.8%	53.2%	89.4%
Glaucoma & Cataract	0.854	0.632	0.808	28.58	26.64	10.5%	34.2%	92.1%
Overall	0.868	0.860	0.891	22.45	19.72	18.1%	51.4%	95.2%

Table 7. Overall and category-wise scleral spur localization performance.

Classification Validation. Following the position validation, we conducted a multi-class classification validation on the ASOCT-MSDC dataset. The primary motivations for this validation are two-fold: first, to verify the dataset's utility for automated multi-disease screening beyond anatomical structure analysis; and second, to establish performance baselines using classic image classification architectures for future comparative research.

We selected two representative deep learning models for this task: ResNet50 and Vision Transformer. *ResNet50* is a seminal convolutional neural network (CNN) that introduced residual learning to mitigate the vanishing gradient problem in deep networks³⁹. By utilizing skip connections, it effectively extracts hierarchical spatial features from AS-OCT scans, which is crucial for identifying complex pathologies. *Vision Transformer* (ViT) applies the Transformer architecture, originally successful in natural language processing, directly to sequences of image patches⁴⁰. Through its self-attention mechanism, ViT captures long-range spatial dependencies across the entire AS-OCT image, providing a global context that complements the local feature extraction of CNNs.

To quantify the performance of the proposed baseline models, we employ several standard classification metrics. For a multi-class classification problem with K classes, the model is optimized using the Cross-Entropy (CE) loss function:

$$\mathcal{L}_{CE} = - \sum_{i=1}^K y_i \log(\hat{y}_i) \quad (6)$$

where y_i is a binary indicator (0 or 1) if class label i is the correct classification for the observation, and $\hat{y}_i$ is the predicted probability for class i .

The classification performance is evaluated using Precision, Recall and the F1-score⁴¹. Due to the fact that our dataset's labels are not balanced, we choose to focus on F1-score rather than accuracy to evaluate the classification performance. These metrics are defined based on True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) for each class:

- **Precision** measures the accuracy of positive predictions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (7)$$

- **Recall** (Sensitivity) measures the proportion of actual positives correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (8)$$

- **F1-Score** is the harmonic mean of Precision and Recall, providing a balanced assessment especially for imbalanced datasets:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (9)$$

For the overall performance, we report the weighted average of these metrics across all four categories in our AS-OCT dataset. The same predefined partitioning strategy used for the segmentation validation was also used for the classification experiments to support fair comparison. The training set was used for gradient-based parameter optimization. The validation set was used only for hyperparameter tuning, early stopping, and check-point selection and was not involved in gradient-based optimization. The test set was kept fixed and was used only for final performance reporting. The classification performance on the test set is summarized in Table 8.

Classification performance differed substantially across diagnostic categories. Normal eyes were classified with the highest performance, likely because they showed regular anterior segment morphology and lacked disease-related lens opacity or angle crowding. Cortical cataract eyes also achieved relatively high F1-scores, reflecting visible lens-related signal changes in many AS-OCT images. In contrast, PACG and PACG-cortical cataract eyes were more difficult to classify. This may be explained by three factors. First, these categories contained fewer samples than the normal and cortical cataract categories, which may reduce the stability of learned decision boundaries. Second, PACG-related AS-OCT changes can be subtle and may overlap with anatomical variation, especially when clinical diagnosis depends on information beyond a single AS-OCT B-scan. Third,

Model / Category	Precision (%)	Recall (%)	F1-Score (%)	Support
ResNet50 (Overall)	85.35	85.03	85.17	167
Normal	100.00	100.00	100.00	50
Cataract	90.62	87.88	89.23	66
Glaucoma	64.00	69.57	66.67	23
Glaucoma&Cataract	64.29	64.29	64.29	28
ViT-Small (Overall)	85.93	84.34	84.91	166
Normal	100.0	100.0	100.0	50
Cataract	94.12	85.33	89.51	75
Glaucoma	57.14	72.73	64.00	22
Glaucoma&Cataract	52.38	55.00	53.66	20

Table 8. Classification Performance on AS-OCT Dataset.

PACG-cortical cataract eyes contain mixed pathological features, and the cataract component may dominate the image appearance, making separation from isolated cortical cataract or isolated PACG more challenging. These results indicate that future classification studies using ASOCT-MSDC should consider class-balanced sampling, cost-sensitive losses, and external validation.

The released image counts are 326 normal, 499 cortical cataract, 143 PACG, and 138 PACG-cortical cataract, making the two PACG-related categories substantially smaller than the cortical cataract category. Future users should apply participant-level stratified splits and class-balanced sampling or class-weighted losses within the training set; oversampling or augmentation should not alter the natural validation and test distributions. Because weighted averages can be dominated by the larger categories, macro-averaged F1-scores and per-class precision, recall, and F1-scores should be reported alongside overall metrics. Repeated or cross-validated analyses may further help quantify the uncertainty of estimates for the smaller PACG-related groups.

Failure-case analysis. Qualitative inspection of the independent test-set predictions revealed that model errors were not randomly distributed, but were closely associated with image quality, anatomical visibility, and disease-related structural changes. For segmentation, the anterior chamber was generally more robust because it is a relatively large and high-contrast region with continuous boundaries. In contrast, iris segmentation was more challenging because the iris is thin, locally low-contrast, and frequently affected by shadowing or angle crowding near the iridocorneal angle. As shown in Fig. 10a, local shadowing can obscure the iris boundary and lead to incomplete or inaccurate iris segmentation. In addition, because the left and right iris were annotated as two separate anatomical labels, some models occasionally confused the two sides. For example, Fig. 10b shows a case in which both the left and right iris regions were incorrectly assigned to the same iris label, resulting in false-positive segmentation on the contralateral side. These errors suggest that left-right iris separation is more sensitive to local contrast, anatomical symmetry, and spatial ambiguity than segmentation of larger structures.

Lens and nucleus-related errors were mainly observed when the posterior lens boundary or the nucleus-cortex interface was weakly visible. This was particularly evident in images with cortical cataract-related signal heterogeneity, where the internal lens texture became uneven and the boundary between the nucleus and surrounding cortex was less distinct. As illustrated in Fig. 10c and d, weak visibility of the nucleus-cortex interface caused inaccurate nucleus delineation, reflecting the difficulty of segmenting internal lens substructures when the anatomical boundary is not clearly defined in AS-OCT. These observations are consistent with the quantitative results, in which structures with clearer and larger boundaries generally achieved better segmentation performance, whereas thin structures or weak internal interfaces showed larger performance variability.

For scleral-spur localization, larger localization errors were mainly observed in PACG and PACG-cortical cataract eyes. In these cases, narrow or closed angles, angle crowding, and disease-related structural deformation may obscure the scleral-spur landmark and reduce the reliability of landmark detection. For disease classification, misclassification occurred more frequently among PACG, cortical cataract, and PACG-cortical cataract categories. This pattern suggests that coexisting pathological features and class imbalance can make image-level classification more difficult, especially when angle closure and lens opacity-related signal changes are simultaneously present. Overall, these failure-case observations indicate that future studies may benefit from improved modeling of low-contrast anatomical boundaries, explicit spatial constraints for left-right iris separation, and additional data from challenging disease subgroups.

Usage Notes

Potential reuse and clinical relevance. ASOCT-MSDC is intended to support the development and benchmarking of automated AS-OCT analysis methods. The anatomical masks can be used to train segmentation models for quantitative anterior segment measurements, including anterior chamber morphology, iris configuration, lens contour, and nucleus-related features. The scleral-spur landmarks provide reference points for angle-related biometric measurements and may facilitate automated assessment of angle configuration. The diagnostic labels allow researchers to evaluate multi-class and multi-task learning approaches in a clinically relevant setting that includes normal eyes, cortical cataract, PACG, and PACG with cortical cataract. The image-quality scores further enable studies on artifact-aware learning, quality-controlled model training, and robustness analysis under clinically realistic imaging conditions.

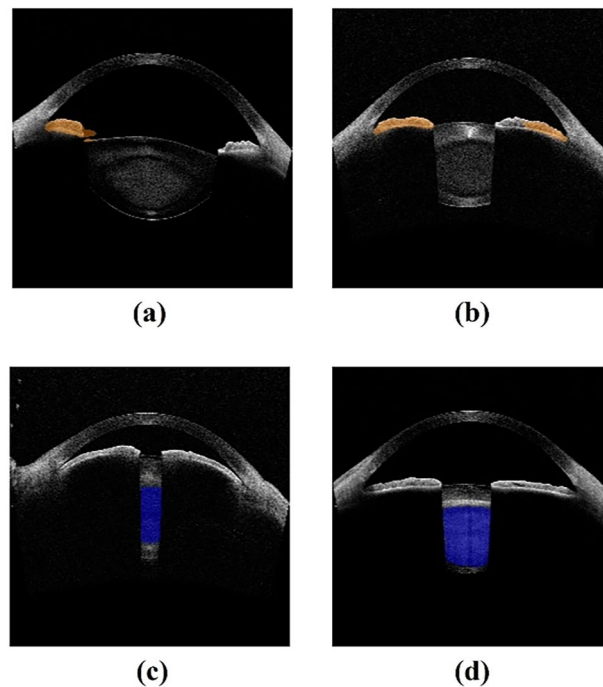


Fig. 10 Representative segmentation failure cases on the independent test set. (a) Mis-segmentation of the left iris due to local shadowing and weak iris contrast. (b) Mis-segmentation caused by left-right iris confusion, where both iris regions were assigned to the same iris label. (c,d) Two examples of nucleus mis-segmentation caused by weak visibility of the nucleus-cortex interface. These examples illustrate that segmentation errors were mainly associated with thin low-contrast structures, left-right anatomical ambiguity, and weak internal lens boundaries.

Limitations. Several limitations should be considered when using ASOCT-MSDC. First, the dataset was collected from a single eye hospital in China. After institutional de-identification and data-minimization procedures during data export, demographic variables such as age and sex were not retained in the final released research database. Therefore, demographic-stratified analyses could not be performed, and external validation is needed before applying models trained on this dataset to other populations. Second, all images were acquired using the Tomey CASIA2 swept-source AS-OCT system. Differences in device hardware, scan protocol, image resolution, wavelength, and vendor-specific preprocessing may lead to domain shift when models are transferred to images acquired using other AS-OCT devices. Direct transfer to platforms such as ANTERION has not been evaluated. Differences in scan geometry, sampling density, intensity scaling, and device-specific image processing may affect both anatomical appearance and pixel-based localization errors, even after image resizing. Models trained with ASOCT-MSDC therefore require external, device-specific validation and calibration before clinical use on another platform. Future multi-device studies could evaluate physical-scale harmonization, device-aware normalization, domain-adaptation methods, and fine-tuning with representative target-device data, with performance reported separately for each device. Third, the diagnostic categories are imbalanced, with cortical cataract eyes being more frequent than PACG and PACG-cortical cataract eyes. This imbalance should be considered when designing training strategies and interpreting classification performance. Fourth, the glaucoma category consists of PACG cases; therefore, findings may not generalize to open-angle glaucoma or other glaucoma subtypes. Fifth, the cataract category represents cortical cataract and does not cover the full spectrum of cataract subtypes. Finally, standardized LOCS III grading records and original slit-lamp photographs were not included in the final released research database, so ASOCT-MSDC should not be used as a LOCS-based cataract severity grading dataset.

Data availability

The released AS-OCT images are de-identified cross-sectional scans and do not contain external facial images or full-view iris photographs. The ASOCT-MSDC dataset is publicly available through figshare at <https://doi.org/10.6084/m9.figshare.31550530>³². The repository contains AS-OCT images, corresponding JSON annotation files, and metadata tables describing image-quality scores and diagnostic categories.

Code availability

The source code for this project, including implementations of nnUNet, TransResUNet, MedSAM, YOLOv11l-pose, ViT and ResNet50 is publicly available at: <https://github.com/Fluorine-Brian/Code-for-AS-OCT-Dataset>.

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Author contributions

Conceptualization: Y.X., J.Z., Y.Y.; Methodology: J.Z., X.F.; Data Acquisition: W.Y., W.C.; Clinical Supervision: W.Y., W.C.; Data Curation: J.Z., Y.Y., X.F.; Manual Annotation: J.D., J.Y., X.Y.; Annotation Review: X.H., X.W., Y.H., D.G., Z.Z.; Experimental Validation: J.Z., Y.Y., X.F.; Manuscript Drafting: J.Z., Y.Y., X.F.; Manuscript Revision: H.F., J.Z., Y.Y., X.F.; Supervision: Y.X., H.F.; Funding Acquisition: Y.X. All authors have reviewed and approved the final version of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to W.C. or W.Y.

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